

PEERLENS

Complete Step-by-Step Instructor Guide for Professors

Welcome to **PeerLens**, an advanced, anonymous peer-evaluation platform designed to simplify group-work analytics, promote collaborative accountability, and mathematically align individual grades with actual team contributions.

This comprehensive manual walks you through setting up your classroom, managing cycles, analyzing data, and grading students.

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1. Core Features & Pedagogical Design

PeerLens is designed around rigorous academic standards to address the standard pain points of group grading:

- **Double-Blind Anonymity:** Students evaluate their peers honestly, knowing that their teammates only see aggregated averages. Individual scores are completely hidden.
- **WebPA Calibration:** Integrates the standard academic *WebPA* algorithm. Rather than giving everyone in a group the exact same project score, PeerLens scales individual grades based on peer-reviewed contribution metrics.
- **Self-Reflection & Calibration:** Students rate themselves alongside their teammates. The system plots self-ratings side-by-side with peer averages to close self-perception gaps.
- **Conflict & Friction Alerts:** Built-in standard deviation analyzers alert the instructor when group members hold conflicting views on individual contributions.
- **Local-First & Sync Ready:** Runs 100% locally in the browser out-of-the-box (no setup needed). For cloud storage and online submissions, it links directly to your free Firebase database.

2. Step 1: Setting Up Your Workspace & Classroom

When you first launch PeerLens as an instructor, you will arrive at the Welcome Screen.

10. Create an Admin Profile:

- In the top-right corner, click the **Profile Selector**.
- Choose **+ New Workspace** to create your own isolated workspace (e.g., "Computer Science Department" or "Spring 2026").
- Workspaces keep your roster lists, rubrics, and settings fully organized and separate.

11. Create Your First Classroom:

- Click **Create Classroom**.
- Enter a course title (e.g., **Algorithms Design - Group A**).
- Click **Add Classroom**. You can create as many classrooms as you need within your workspace.

3. Step 2: Onboarding & Managing Student Rosters

Go to the **Enrollment & Teams** tab to manage your students.

Option A: The Import Enrollment Wizard (Fastest)

12. Click **Launch Roster Onboarding Wizard**.

13. Select your data import method:

- **XLSX / CSV File:** Upload your enrollment sheet directly.
- **PDF Document:** Upload a student roster PDF.
- **Excel / Google Sheets Copy-Paste:** Copy your spreadsheet columns and paste them directly into the text field.

14. **Map Your Columns:** The wizard will automatically guess your columns. Ensure that **Full Name** and **Email ID** are mapped correctly (Unique Student ID, Team Name, University, and Country are optional).
15. **Verify & Edit:** The wizard opens an interactive draft spreadsheet. Any duplicate emails or missing IDs are highlighted in red. You can edit cells, add rows, or delete rows directly in the wizard before finalizing.
16. Click **Finalize Import**.

Option B: Manual Add

- Under **Quick Actions**, click **Add Member Manually**.
- Fill in the Student ID, Name, Email, and Team Name (if they are already grouped).

Option C: Auto-Partitioning Teams

- If you have imported your students but haven't grouped them into teams, go to **Quick Actions** and click **Auto-Partition Balanced Teams**.
- Enter your target team size (e.g., 3 or 4 students per group), and PeerLens will automatically shuffle the roster and distribute them into balanced, numbered teams.

4. Step 3: Building Your Evaluation Rubric

Go to the **Evaluation Rubric** tab to specify what metrics your students will use to rate each other.

17. Customize Criteria:

- You can edit the criteria names (e.g., *Technical Reliability*, *Communication Quality*, *Problem Solving Contribution*).
- Define the score boundaries (e.g., a scale from **1** to **10**, or **1** to **5**).
- Assign a **Weight** to each criterion. If *Technical Quality* is twice as important as *Punctuality*, set its weight to **2** and punctuality to **1**.

18. Add or Remove Criteria:

- Click **+ Add Rubric Metric** to add a new assessment point.
- Click the **Trash** icon next to any metric to remove it.

19. Save Evaluation Deadline:

- In the sidebar card, set a **Submission Deadline Date & Time**.
- Once this deadline is reached, the student evaluation forms will automatically lock down, preventing late entries.

5. Step 4: Dispatching Invites & Tracking Submissions

To enable students to grade each other, they require their secure, personal access links. Go to the **Notification Center** tab.

1. Select Your Active Mailing Mode

- **Demo Mode — No emails sent:** Free mode. Generates the list of personal student links directly on your screen. You can copy them manually or send them via your own email/LMS system.
- **EmailJS Adapter:** Links to your free EmailJS account to send emails directly from the browser.
- **Brevo API Service:** Integrates with Brevo SMTP API (provides 300 free emails/day) to dispatch high-quality, professional email templates.

2. Send Invitation Campaign

- Click **Send Links to All**.
- The dispatcher will run, showing a real-time progress bar and log console. Each student will receive a secure email containing their personal grading link.
- **Important:** Students do *not* need to register an account. Clicking their personal link securely opens their personalized evaluation portal immediately.

3. Send Reminders

- As the deadline approaches, go to the **Notification Center** and click **Send Reminders**.
- This will *only* send emails to students who have **not yet submitted** their evaluations, leaving active submitters uninterrupted.

4. Monitor Progress

- You can check submission progress on the main dashboard cards under **Submission Rate**.
- In the student grid, a green **Submitted** checkmark will appear next to each student who has successfully filled out their form.

6. Step 5: Grade Analytics & The WebPA Calibration Simulator

Once submissions start arriving, navigate to the **Grade Analytics** tab.

The Real-time Calculation Matrix Table

This table compiles all anonymous student ratings and shows:

- **Peer Average (Raw):** The raw average score given to the student by their teammates.
- **Self Score:** The score the student gave themselves (used for calibration, excluded from calculations).
- **Teammate Consensus (Standard Deviation):**
- **Low Std Dev (Green):** Teammates are highly aligned on this student's contributions.
- **High Std Dev (Red Alert):** Signifies conflict. Teammates gave widely different ratings, indicating friction in the group that the instructor should investigate.
- **WebPA Factor:** The individual's contribution ratio relative to the team's average. A factor of **1.0** means they contributed exactly their share. A factor of **1.15** means they carried the group. A factor of **0.70** means they underperformed.

WebPA Grade Calibration Simulator

This sidebar simulator converts group project grades into individual scores:

20. Enter your **Project Base Mark** (the grade you gave the team's overall deliverable, e.g., **85** out of 100).
21. Adjust the **Fudge Weight Slider** (scaling factor weight):
 - **At 0% Weight:** The scaling factor is disabled. Every student gets the identical team grade of **85**.
 - **At 50% Weight (Balanced):** Individual grades scale gently. High contributors might get **88**, and lower contributors **81**.
 - **At 100% Weight (High Penalty/Reward):** Grades scale aggressively. The high contributor gets **94**, while the low contributor drops to **68**.
22. Click **Download Grades Sheet** to download a compiled CSV spreadsheet compatible with Excel, Google Sheets, or direct upload to your LMS (Canvas, Blackboard, Moodle).

7. Step 6: Transitioning Between Cycles / Milestones

If your course has multiple group deliverables (e.g., Midterm Project, Final Release), you can track progress across review rounds.

23. Archive the Current Milestone:

- On the **Evaluation Rubric** tab, locate the **Milestone Archiving Control** card.
- Type in a name (e.g., **Midterm Evaluation**).
- Click **Archive Current Milestone & Restart**.

24. What this does:

- Saves all current evaluations, peer averages, and comments into a historical milestone database.
- Resets the active review cards to "Pending" and clears the submissions list so students can evaluate each other fresh for the next cycle.

25. Reviewing History:

- Students and instructors can view historical milestones in their dashboards to track performance growth curves across the semester.
- You can delete old milestones in the sidebar controls if needed.

8. Student Portal: What Your Students See

When a student clicks their secure link, their customized **Student Portal** opens.

Phase 1: Review Submission (Before submitting)

- The portal lists all of their teammates (excluding themselves) and a section for self-evaluation.
- **Numerical Grades:** For each teammate, they rate them across the criteria. They can quickly click the **Achievement Tiers** (*Below Expectations, Progressing, Meets Expectations, Exceeds Expectations, Distinguished Performer*) to snap the sliders to pedagogical midpoints.
- **Praise Tags:** They select custom recognition chips for their teammates (e.g., *Super Supportive, Great Communicator, Always Punctual*).
- **Written Feedback:** They write concise blocks specifying:

26. *What were this teammate's key strengths?*

27. *What is one area where this teammate can grow?*

- Click **Submit Anonymous Feedback**.

Phase 2: Performance Dashboard (After submitting)

Once a student submits, their portal transforms into a personal visual report:

- **Completion Tracking:** Shows how many teammates have submitted. Their report averages unlock as soon as reviews are received.
- **Overall Rating Dial:** A circular chart displaying their overall teammate evaluation percentage.
- **Self vs. Peer Comparison Graph:** A side-by-side bar chart plotting their self-rating against teammate averages for each metric. Excellent for addressing self-perception gaps.
- **Peer Recognition Tag Cloud:** Displays a tag cloud of the praise chips their teammates gave them (with aggregate counts).
- **Achievement Badges Hub:** Gamified badges unlock based on their high scores (e.g., *Dependable Anchor* for consistency, *Creative Catalyst* for ideas).
- **Shuffled Qualitative Comments:** Displays all strengths and growth comments left by their teammates. **Highly Secure:** The comments are completely shuffled and stripped of names, protecting teammate anonymity.
- **Milestone Progress Charts:** Line charts tracing their progression curve across different milestones.

9. Advanced: Configuring Cloud Sync & Firebase

By default, PeerLens runs 100% locally in your browser's **localStorage** (highly secure and offline-mode compatible). To sync data to the cloud so students can access their portals online:

28. Create a free account on [Firebase Console](https://console.firebase.google.com/).
29. Create a new project and add a **Web App** to it.
30. Go to **Firestore Database** and click **Create Database** (start in Test Mode or configure security rules).
31. Copy the Web App configuration keys:

```
{
  "apiKey": "your-api-key",
  "authDomain": "your-project-id.firebaseio.com",
  "projectId": "your-project-id",
  "appId": "your-app-id"
}
```

32. In **PeerLens**, go to the **Data & Sync** tab.
33. Paste the keys into the inputs and click **Activate Cloud Storage**.
34. The indicator in the top header will change to green **Cloud Connected**, indicating all classrooms are securely synchronized live in the cloud database!